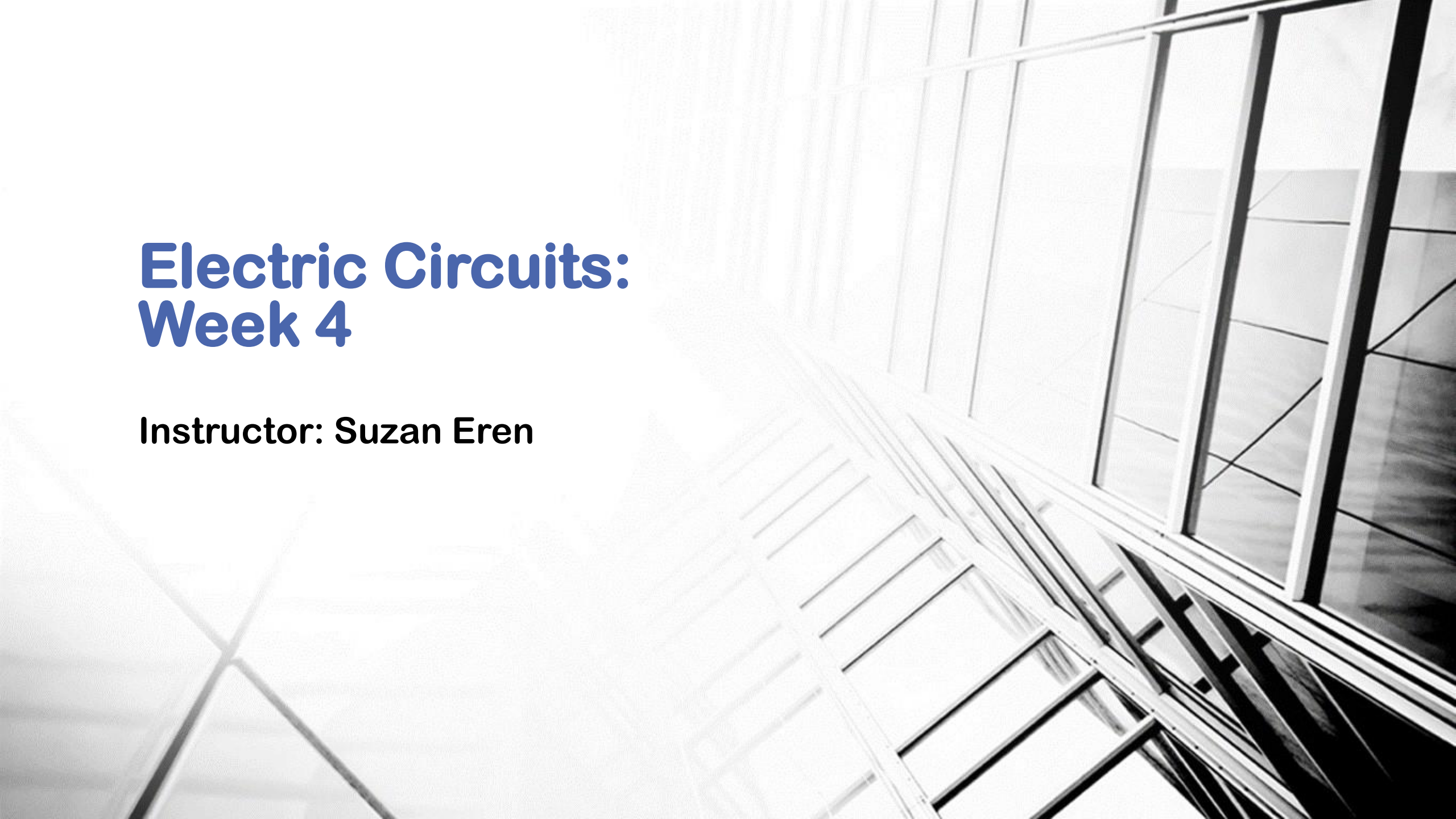
Electric Circuits: Week 4

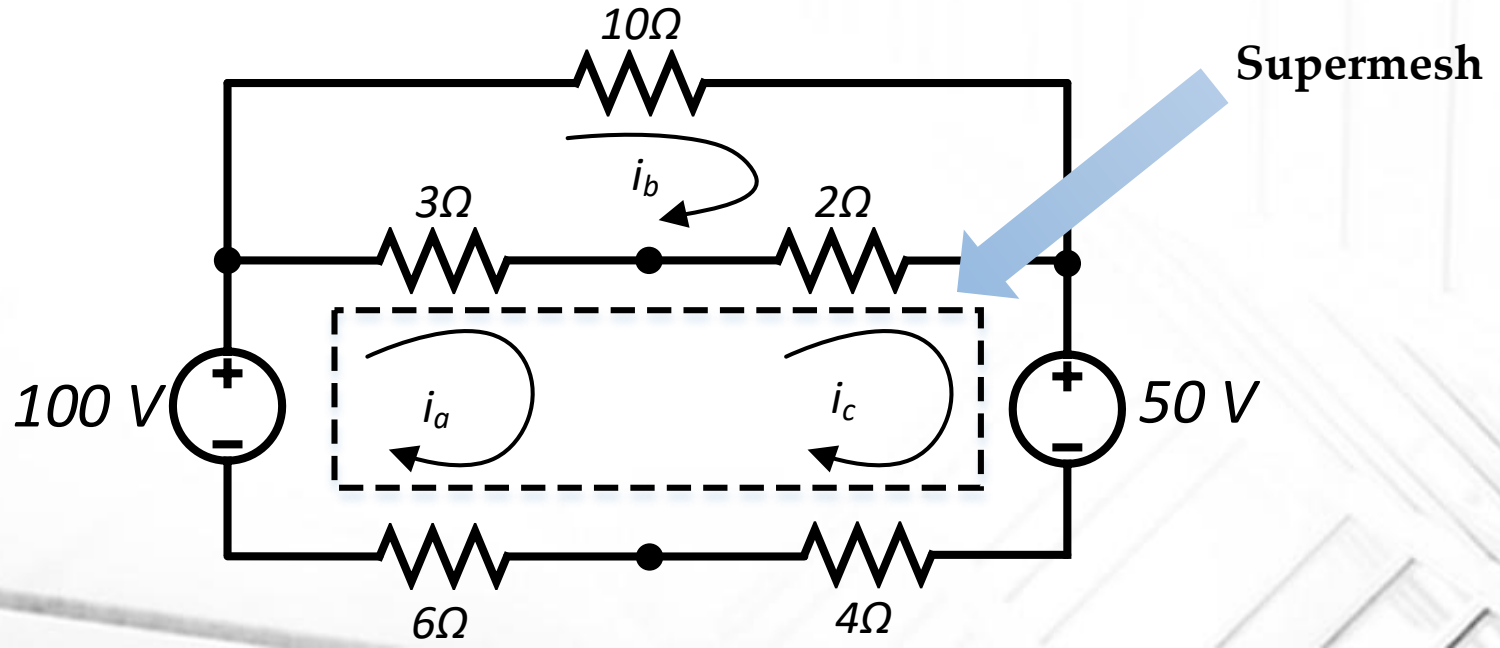
Instructor: Suzan Eren



Mesh-Current Method: Special Case

Supermesh: When there is a current source between two meshes, we can remove it to create a supermesh. The voltages around the supermesh are expressed in terms of the original mesh currents. This is a simpler way to approach this special case.

Example: Let's solve the previous example using a supermesh.



Node Voltage vs. Mesh Current Method

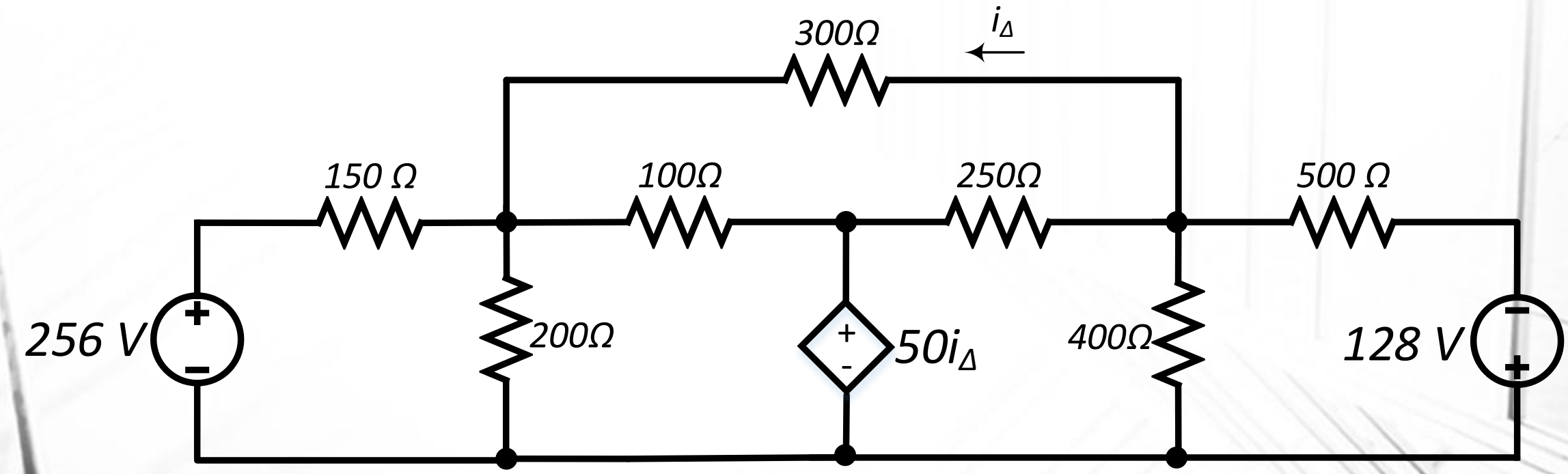
How to determine: Which of the methods will require fewer equations?

- Are there any supernodes? Node-Voltage Method may help reduce the number of equations
- Are there any supermeshes? Mesh-Current Method may help reduce the number of equations

Let's look at two examples to help us determine how to pick the correct method…

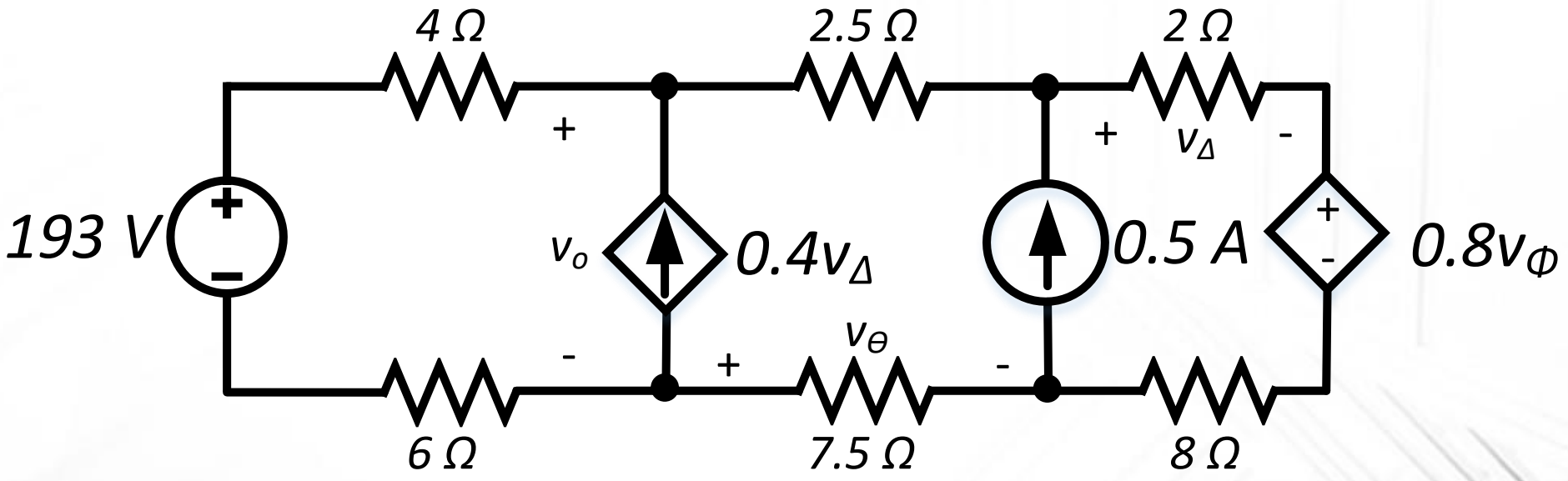
Node Voltage vs. Mesh Current Method

Example 1: Which method is better for the following circuit? Find the power dissipated in the 300Ω resistor.



Node Voltage vs. Mesh Current Method

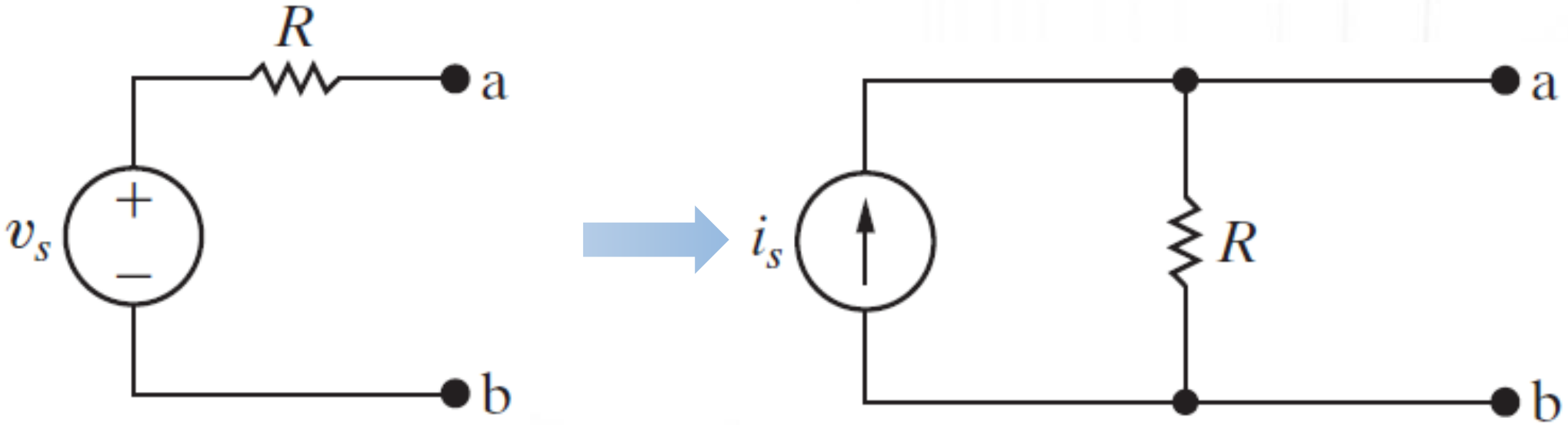
Example 2: Which method is better for the following circuit? Find the voltage v_o.



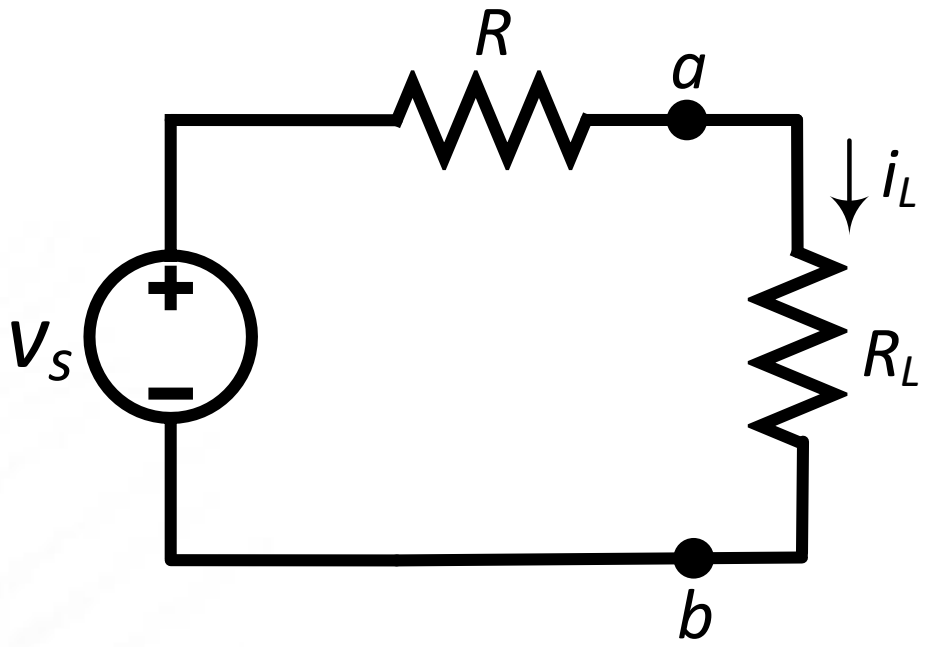
Source Transformation

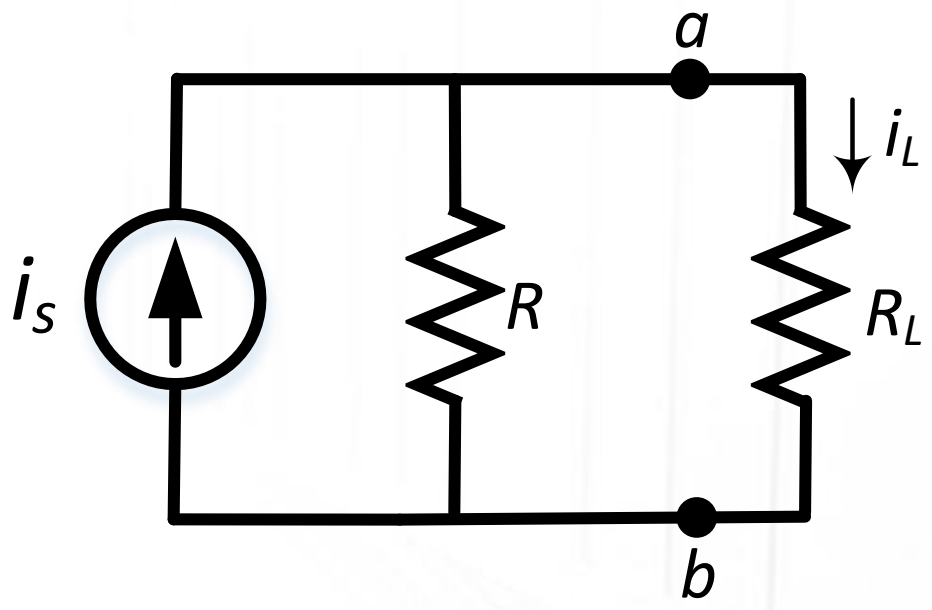
Now that we've learned several methods to solve circuits, how do we simplify the process? One idea: Source Transformations

- Voltage across terminals a and b, and current through terminals a and b should be the same for both cases



Source Transformation



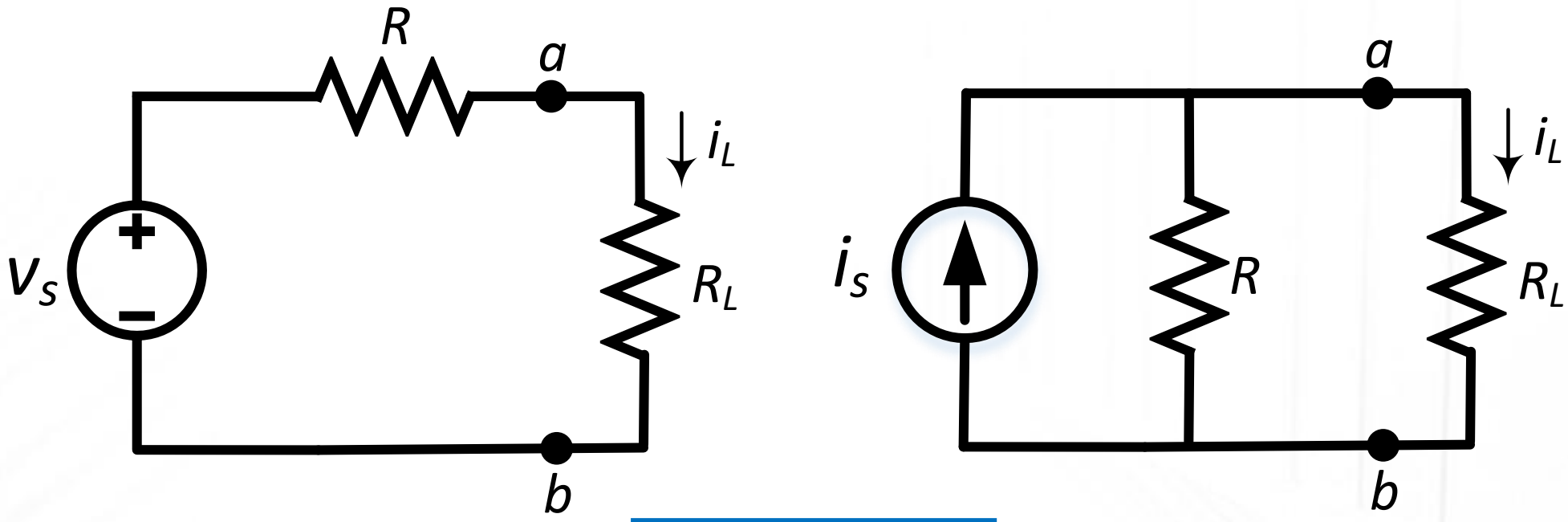


What happens if we equate these two expressions?

$$i_L = \frac{v_s}{R + R_L}$$

$$i_L = \frac{R}{R + R_L} i_s$$

Source Transformation

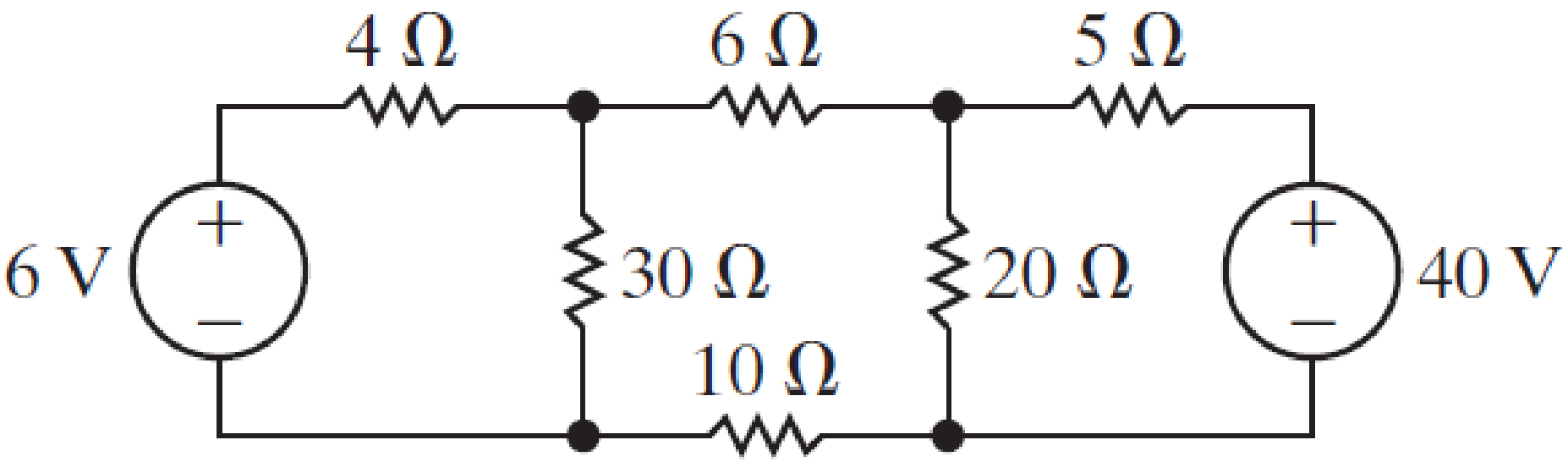


We can transform a voltage source to a current source (and vice versa) using this equation.

$$i_s = \frac{v_s}{R}$$

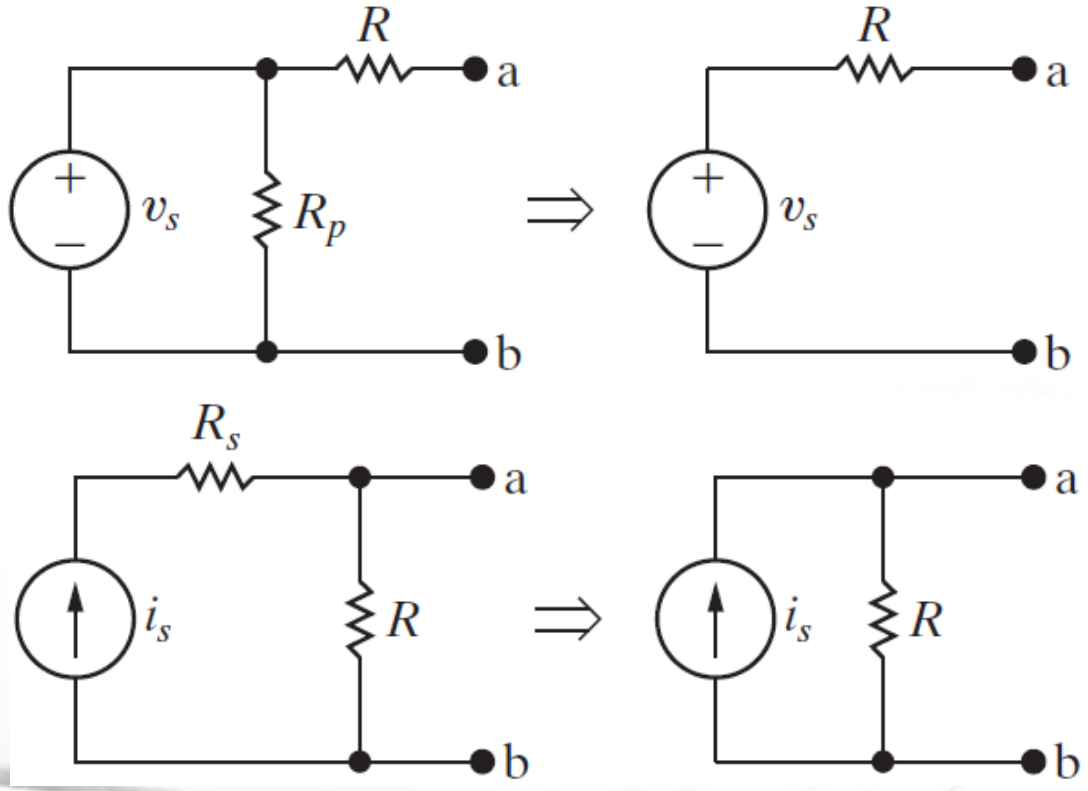
Source Transformation

Example: What is the power associated with the 6V source, and is the power being absorbed or delivered?

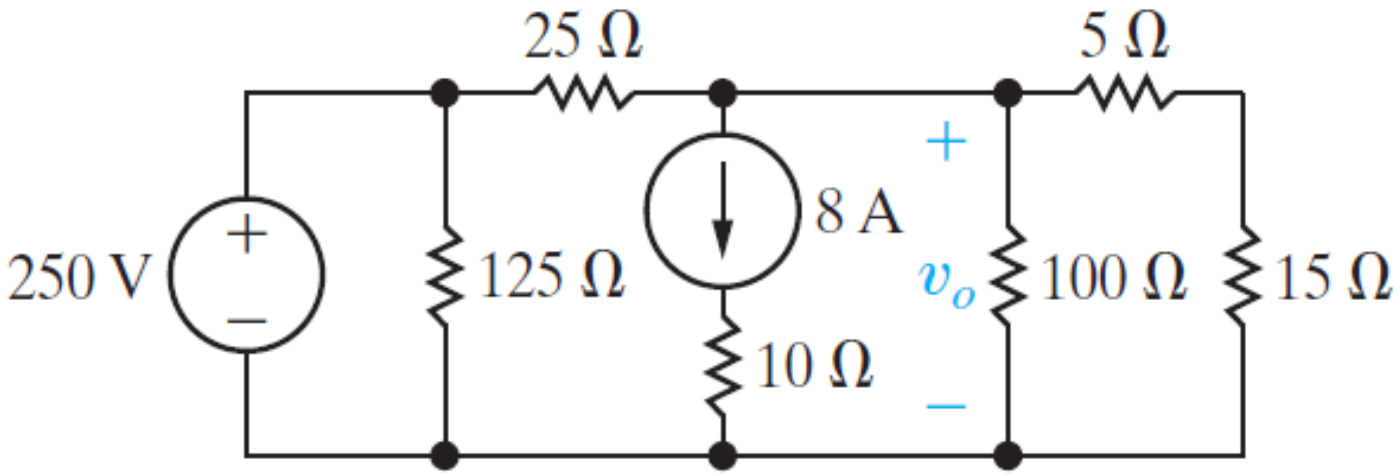


Source Transformation: Special Cases

What if there is a resistance in parallel with a voltage source, or a resistance in series with a current source? It has no effect on the equivalent circuit (but does have an effect on power calculations).



Source Transformation: Special Cases



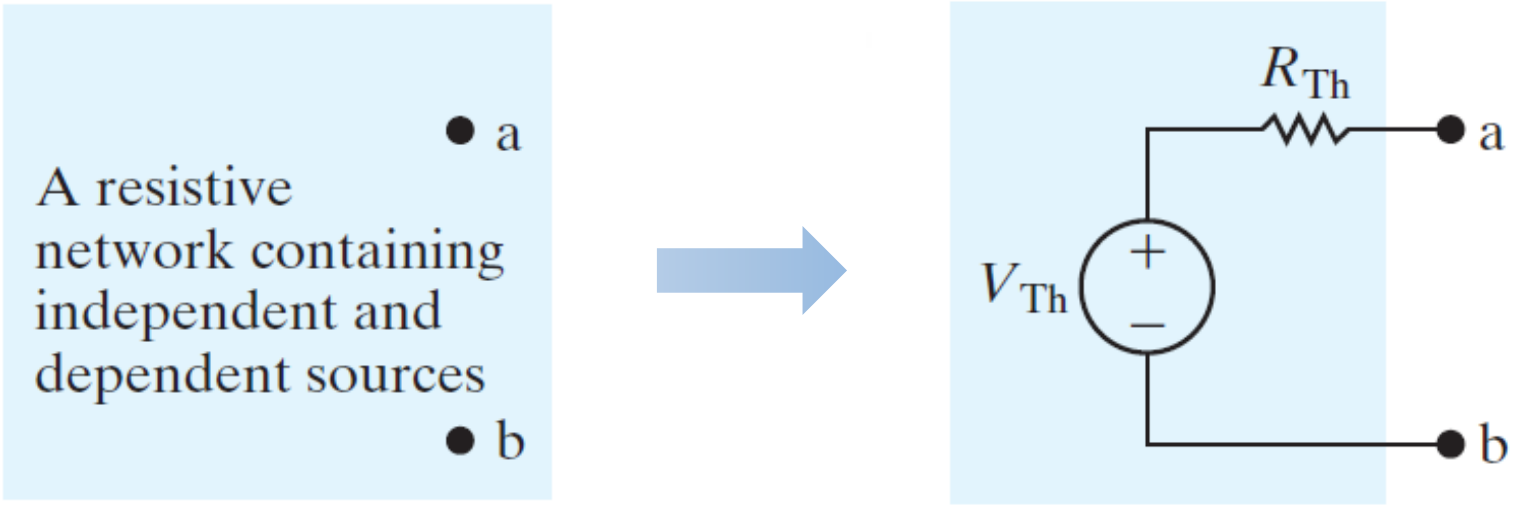
(a) Use source transformations to find the voltage v_o in the circuit shown.

(b) Find the power developed by the 250 V voltage source.

(c) Find the power developed by the 8 A current source.

Thévenin Equivalent Circuit

What if we are only interested in the voltage and current at the terminals a and b? We can simplify the rest of the circuit by finding its Thévenin equivalent circuit…

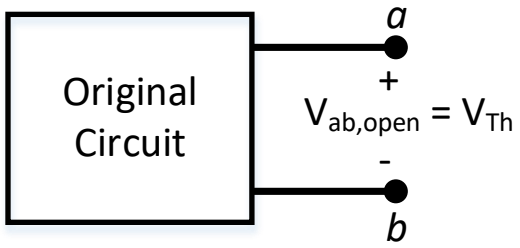


The Thévenin equivalent circuit can be used to represent any circuit made of linear elements

Thévenin Equivalent Circuit

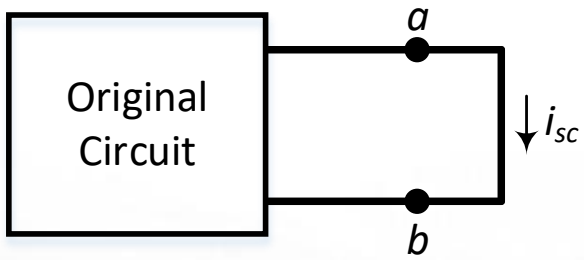
To find the Thévenin equivalent circuit, follow this procedure:

1. Calculate the open-circuit voltage in the original circuit.



$$v_{ab} = V_{Th}$$

2. Find the short circuit current.

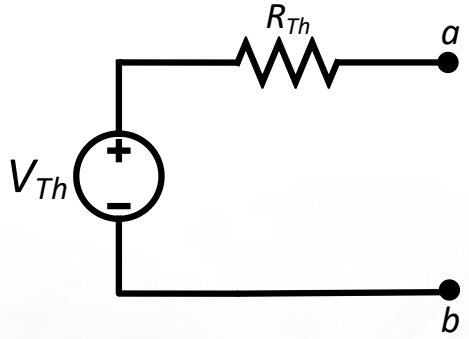


Thévenin Equivalent Circuit

3. The following relationship allows us to solve for R_{Th}.

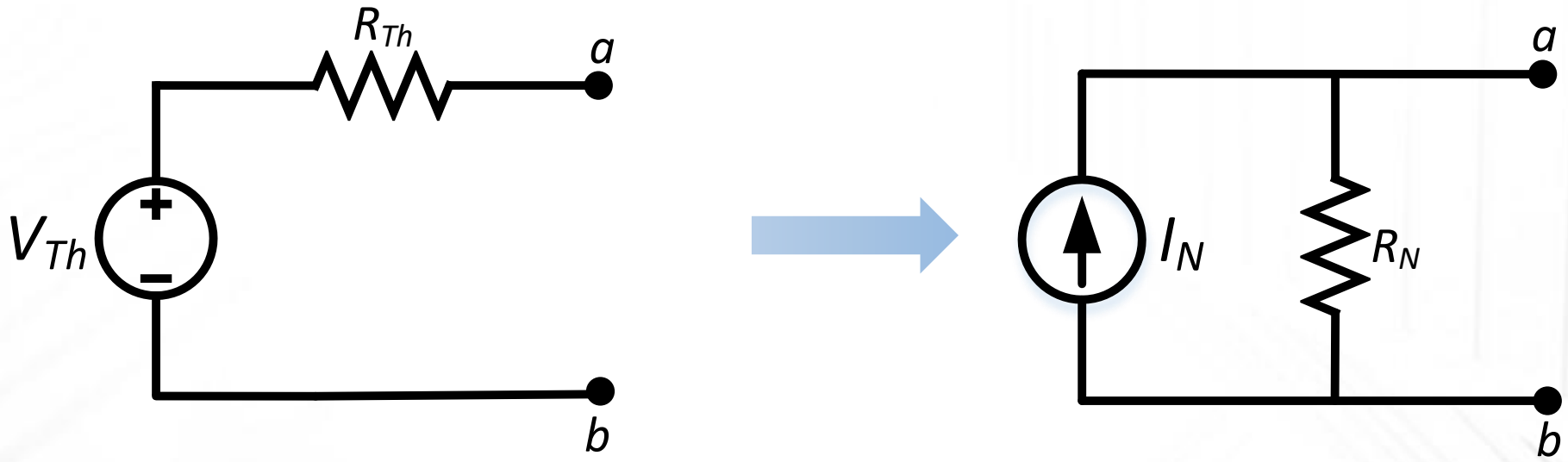
$$i_{sc} = \frac{V_{Th}}{R_{Th}} \quad \Rightarrow \quad \boxed{R_{Th} = \frac{V_{Th}}{i_{sc}}}$$

4. Now we have V_{Th} and R_{Th}, so we have found our Thévenin equivalent circuit.



Norton Equivalent Circuit

The Norton equivalent circuit is the source transformation of the Thévenin equivalent circuit.

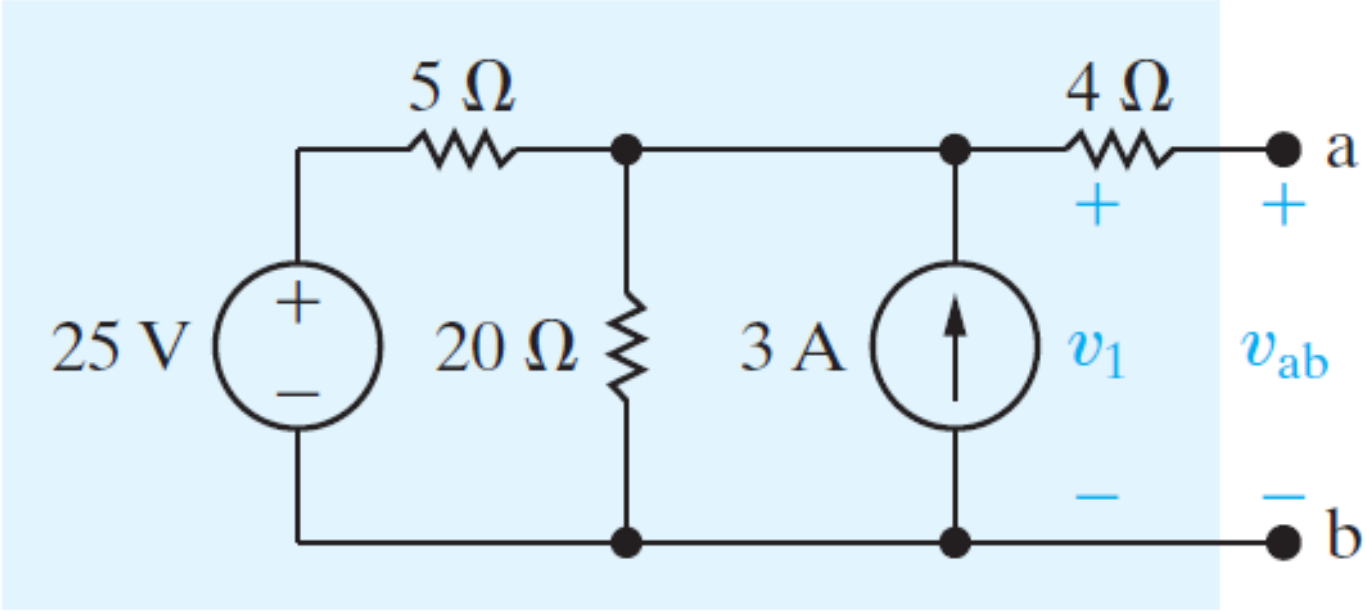


$$I_N = i_{sc}$$

$$R_N = R_{Th}$$

Thévenin Equivalent Circuit

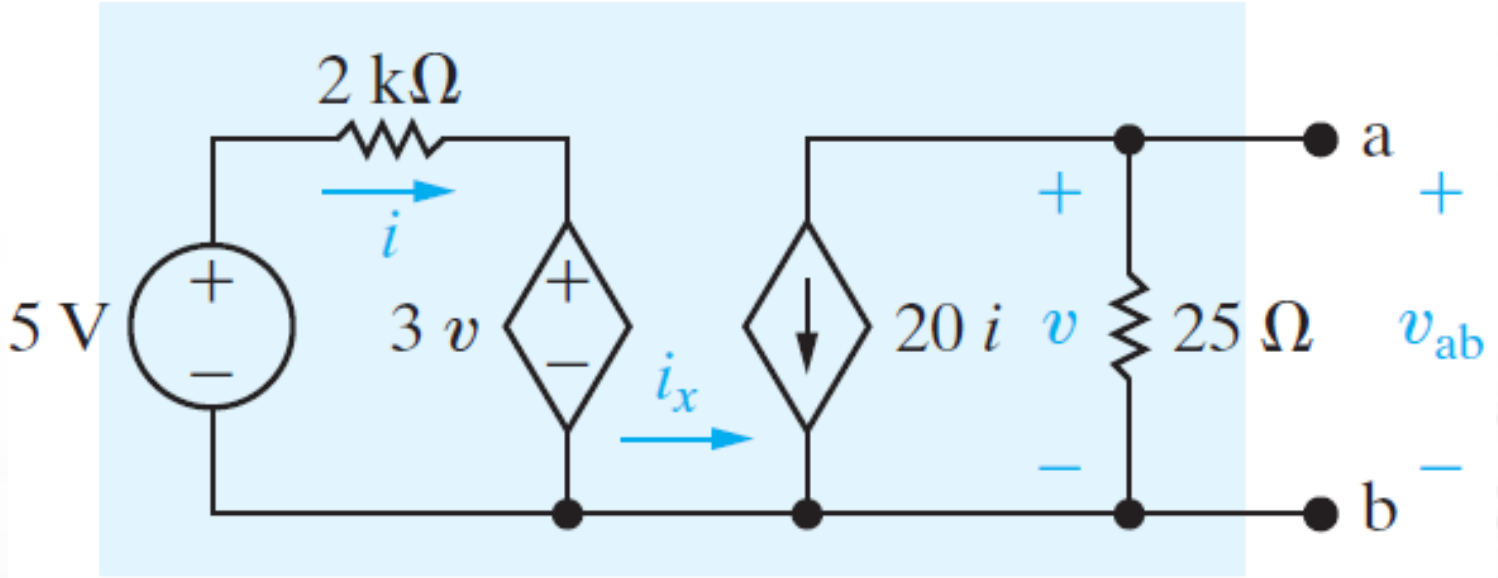
Example: Find the Thévenin and Norton equivalent circuits.



Circuits with Dependent Sources

Dependent sources are treated differently than independent sources, because they are reliant on other voltages and currents in the circuit.

Example: Find the Thévenin equivalent circuit of a circuit with a dependent source.

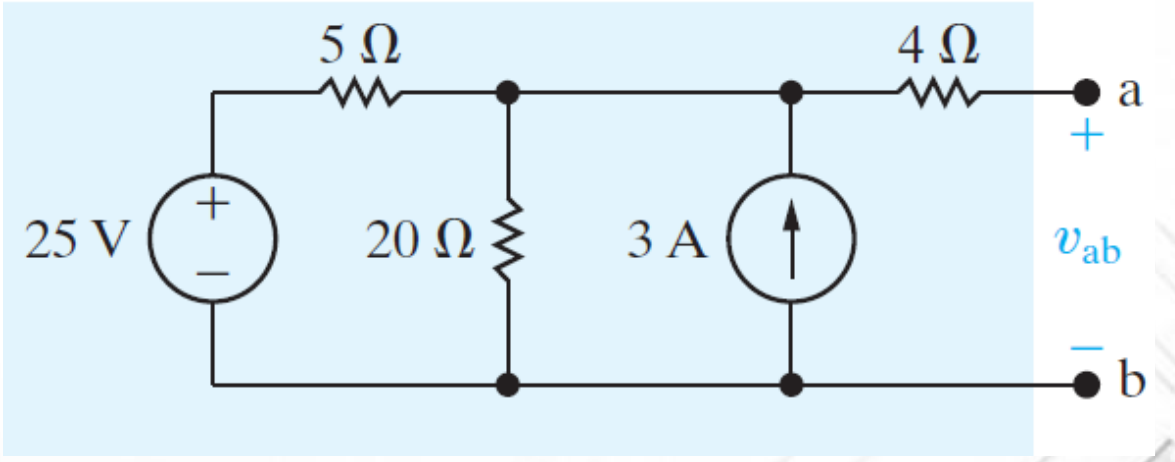


Determining R_{Th}

When the circuit contains **only independent sources**, there is an easier way to determine R_{Th}.

1. Deactivate all independent sources: replace voltage sources with short circuits, and replace current sources with open circuits.
2. Determine the resistance R_{Th} at the designated terminal pair.

Example: Find R_{th}.



Determining R_{Th}

When the circuit contains a combination of **independent and dependent sources**, there is another method that can be used to determine R_{Th}.

1. Deactivate all independent sources: replace voltage sources with short circuits, and replace current sources with open circuits.
2. Apply a test voltage source or test current source to the Thevenin terminals a,b.
3. Determine R_{th}.

$$R_{Th} = \frac{v_T}{i_T}$$

Determining R_{Th}

Example: Find R_{th} by using a test source.

